MIL-P-63159(AR)
25 June 1982
Superseding
(see 6.8)

MILITARY SPECIFICATION

PROPELLANT GRAIN
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for rocket motor bodies used on M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-60	Tape, Packing, Water Proof
PPP-T-76	Tape, Pressure Sensitive Adhesive paper, Water Resistant (for carton sealing)

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1110

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MILITARY

MIL-B-117	Bag, Sleeve and Tubing - Interior Packaging
MIL-C-13609	Casting Solvents for OIO, OGK, and AHH Propellants
MIL-A-48078	Ammunition, Standard Quality Assurance Provision
MIL-P-60432	Propellant Powder, Base Grain, AHH

STANDARDS

MILITARY

MIL-STD-286	Propellants, Solid Sampling, Examination and Testing
MIL-STD-453	Inspection, Radiographic
MIL-STD-652	Propellants, Solid, for Cannons, Requirements and Packing
MIL-STD-1167	Ammunition Data Cards

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.9)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

11828869 Propellant Grain

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Casting powder. The casting powder shall comply with the requirements of MIL-P-60432 (see 6.4). Except for the requirements for total volatiles and residual solvent content this specification shall be advisory for a contractor that produces the propellant grain.

3.1.2 Casting solvent. The casting solvent shall comply with the requirements of MIL-C-13609, Type III. This requirement shall be advisory for a contractor that produces the propellant grain except for the Nitroglycerin content which shall be maintained at ± .10% for each propellant grain lot.

3.2 Propellant grain. The inhibited propellant grain shall comply with all requirements specified on drawing 11828869, with all requirements specified herein and with all requirements specified in applicable specifications.

3.3 Composition. The propellant grain composition shall be as specified in Table I when determined as specified in 4.5.1.

TABLE I Composition

Ingredient	Percent
Nitrocellulose	53.9 Min
Nitroglycerin	31.6 Nominal
Triacetin	8.8 Nominal
2-Nitrodiphenylamine	0.9 Minimum
Lead Salicylate	1.2 Nominal
Lead 2-Ethyl Hexoate	2.0 Nominal

3.4 Heat of explosion. The heat of explosion shall be no less than 940 calories per gram and no more than 970 calories per gram when determined as specified in 4.5.5.

3.5 Strand burning rate. The propellant shall meet the burning rate specified in Table II when determined as specified in 4.5.4.

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TABLE II. Burning rate

Temperature (°F)	Pressure (P.S.I)	Burning Rate (Inches/Seconds)
-40	600	0.359 Minimum
+140	1600	0.581 Maximum
+70	600	Informational
+70	700	Informational
+70	800	Informational
+70	900	Informational
+70	1000	Informational
+70	1100	Informational
+70	1200	Informational

3.6 Cracks, foreign material and porosity. The finish machined propellant grain shall be free of cracks, foreign material and porosity as defined herein. When required, radiographic inspection for defects shall be performed in accordance with 4.5.6.

3.6.1 Radiographic examination:

a. Cracks. There shall be no cracks (shown on the radiograph as dark lines) in excess of .06 inches in length.

b. Foreign Material. There shall be no foreign material (shown by bright spots on the radiograph) in excess of 0.06 inches in any shape or dimension.

c. Porosity. There shall be no porous areas (shown by low density or rounded shadows on radiographs) in excess of .03 square inches individually or with a maximum combined area of .06 square inches.

3.6.2 Visual examination.

a. Cracks. There shall be no cracks exceeding .06 inches in length.

b. Foreign Material. There shall be no foreign material exceeding .06 inches in any shape or dimension.

c. Porosity. There shall be no porous areas exceeding .03 square inches individually or a maximum combined area of .06 square inches.

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3.7 Inhibitor separation. The propellant grain shall contain no individual separations between the inhibitor and the grain (delamination) greater than 0.01 square inches. The total combined area of propellant inhibitor separations shall be no greater than 0.02 square inch. There shall be no propellant inhibitor separation within one inch of any split surface.

3.8 Taliani test. The rate of evolution of gas from samples subjected to the test specified in 4.5.2 shall be such that the characteristics of a curve showing rate of evolution (millimeters of mercury versus time) meets the following criteria:

Slope of the curve at 100mm of mercury shall be less than 1.00.
Slope of the curve at 100 minutes shall be less than 0.90.
Time to 100mm of mercury shall be more than 100 minutes.

3.9 Physical properties. Compression load and deformation.

3.9.1 High temperature. The compression readings from each casting shall be no less than 870 pounds, and the deformation readings from each casting shall be no less than 53 percent at 140 $\pm$ 5°F when tested as specified in 4.5.3.

3.9.2 Low temperature. The compression readings from each casting shall be no less than 6990 pounds and the deformation readings from each casting shall be no less than 5.25 percent at -40 $\pm$ 5 °F when tested as specified in 4.5.3.

3.10 Density. The density of the propellant grain shall be no less than .057 pounds per cubic inch at 77 $\pm$ 5 °F (21° $\pm$ 3°C) when determined as specified in 4.5.7.

3.11 Data feedback. Production data feedback (including serial numbers) for each sublot or skid shall be provided in the contractors own format for the inspections and tests specified in 4.4.2.1 thru 4.4.2.3 (see 6.5).

3.12 Data cards. Serialized ammunition data cards shall be provided for each inspection sublot or skid in accordance with MIL-STD-1167.

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3.13 Surveillance sample. Surveillance samples from each inspection sublot or skid of propellant grain shall be forwarded to the acquisition agency (see 6.6).

3.14 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.15 Workmanship.

3.15.1 Propellant, grain, assembly. The finish machined propellant grain assembly shall be free of dirt, grease, chips, gouges, solvent drippings, and other defects which would make the item unsuitable for its intended use.

3.15.2 Inhibitors. The inhibitors (circumferential and disc) shall be free from dirt, grease, chips, gouges, cracks, and other defects which would make the item unsuitable for its intended use.

4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

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4.3.1 Submission. When the contract requires first article inspection (see 3.14), the contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Description	Drawing	Quantity required
Propellant castings	---	2
Propellant grains (unsplit)	---	7
Propellant grains (split)	11828869	7

4.3.2 Inspection to be performed. See MIL-A-48078 and the first article inspection table herein.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63159 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.3.2.1	Propellant, Grain		1 OF 1	11828869
				NEXT HIGHER ASSEMBLY
				9299057

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Grain casting tests	2		3.2, 3.3 3.4, 3.5 3.8, 3.9 3.10, 3.15	4.4.2.1
	Propellant grain (unsplit)	7		3.2,3.15	4.4.2.2
	Propellant grain (split)	7		3.2,3.6, 3.7, 3.11, 3.15	4.4.2.3

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots of propellant grain shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot shall consist of the quantity of finished grain produced from a single lot of base grain or casting powder required to produce the entire procurement quantity specified in the contract; and an inspection sublot or skid shall consist of castings produced from a single lot of base grain or casting powder and a single batch or dessicator of casting solvent. The lot interface number shall be changed with each batch or dessicator of casting solvent.

4.4.2 Examination. See MIL-A-48078 and the quality conformance inspection tables herein.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Grain Casting Tests	1 OF 1	11828869
			NEXT HIGHER ASSEMBLY 9299057

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Thickness of inhibitor cylinder (prior to casting)		100%	3.2	Gage
102.	Chemical composition	(a)	--	3.3	4.5.1
103.	Taliani test	(b)	--	3.8	4.5.2
104.	Heat of explosion	(c)	--	3.4	4.5.5
105.	Strand burning rate	(d)	--	3.5	4.5.4
106.	Density	(e)	--	3.10	4.5.7
107.	Physical properties at 140°F	(f)	--	3.9.1	4.5.3
108.	Physical properties at -40°F	(g)	--	3.9.2	4.5.3
Minor					
201.	Evidence of poor workmanship		100%	3.15	Visual

NOTES:

(a) See 4.4.3.1.1 (c) See 4.4.3.1.3 (e) See 4.4.3.1.5 (g) See 4.4.3.2.2
(b) See 4.4.3.1.2 (d) See 4.4.3.1.4 (f) See 4.4.3.2.1

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Propellant Grain, Prior to Splitting	1 OF 1	11828869
			NEXT HIGHER ASSEMBLY 9299057

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Total length		100%	3.2	Gage
102.	Outside diameter of inhibitor		100%	3.2	Gage
103.	True position of inside diameter to outside diameter		100%	3.2	Gage
104.	Width of cut		1/8 hrs.	3.2	Gage/4.4.4
Minor					
201.	Inside diameter		100%	3.2	Gage
202.	Diameter of chamfer on inside diameter of inhibited end		100%	3.2	Gage
203.	Diameter of chamfer on inside diameter on uninhibited end		100%	3.2	Gage
204.	Chamfer on outside diameter missing or improper		100%	3.2	Visual
205.	Evidence of poor workmanship		100%	3.15	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

MIL-P-63159 (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Propellant Grain, After Splitting				11828869
					NEXT HIGHER ASSEMBLY 9299057
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Excessive cracks, foreign material or porosity		100%	3.6.2	Visual
102.	Exposed propellant at intersections of inhibitor surfaces		100%	3.2	Visual
103.	Inhibitor separation		100%	3.7	Visual
104.	Surface finish of saw cut surface		100%	3.2	Visual
105.	Finished weight of grain		100%	3.2	Balance
106.	Missing spacer or incorrect spacer material		100%	3.2	Visual
107.	Grain X-Ray		100%	3.6.1	Visual/X-Ray/4.5.6
108.	Web variation		100%	3.2	Gage
109.	Coded grain identification		100%	3.2	Visual
Minor					
201.	Marking, missing, or improper		100%	3.2	Visual
202.	Evidence of poor workmanship		100%	3.15	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63159 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.4	Packing and Marking	1	1	See 5

NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101.	Grains not individually packed in chip-board tubes		100%	5.1	Visual
102.	Drum or barrier bags not as prescribed		100%	5.1	Visual
103.	Protective bag missing or damaged		100%	5.1	Visual
104.	Tape missing or improperly applied		100%	5.1	Visual
105.	Layers not separated by separator		100%	5.1	Visual
106.	Filler material does not provide tight pack		100%	5.1	Visual
107.	Drum split or damaged		100%	5.1	Visual
108.	Drum not properly and securely sealed		100%	5.1	Visual
109.	Surveillance sample not forwarded	(a)		3.13	Visual
110.	Data cards not provided	(a)		3.12	Visual
Minor					
201.	Marking missing or improper		100%	5.2	Visual
202.	Evidence of poor workmanship		100%	5.1	Visual

NOTES:

(a) 1 per skid or sublot

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing.

4.4.3.1 Chemical properties.

4.4.3.1.1 Chemical composition. A random sample of grain shall be selected whenever grains are cast from a new batch or dessicator of casting solvent and tested in accordance with 4.5.1. If any sample fails to meet the requirements of 3.3, the sublot(s) or skids represented by the sample shall be rejected.

4.4.3.1.2 Taliani test. A random sample of grain shall be selected whenever grains are cast from a new batch or dessicator of casting solvent and tested in accordance with 4.5.2. If any sample fails to meet the requirements of 3.8, the sublot(s) or skids represented by the sample shall be rejected.

4.4.3.1.3 Heat of explosion. A random sample of grain shall be selected whenever grains are cast from a new batch or dessicator of casting solvent and tested in accordance with 4.5.5. If any sample fails to meet the requirements of 3.4, the sublot(s) or skids represented by the sample shall be rejected.

4.4.3.1.4 Strand burning rate. Twenty seven strands shall be formed from a random sample of grain selected whenever grains are cast from a new batch or dessicator of casting solvent and tested in accordance with 4.5.4. Three strands shall be used to determine the average burning rate at each temperature-pressure combination specified in 3.5. If any of the average burning rates fail to meet the applicable requirements of 3.5, the sublot(s) or skids represented by the sample shall be rejected.

4.4.3.1.5 Density. A sample shall be selected from the center of each casting in the sublot or skid and tested in accordance with 4.5.7. If any sample fails to meet the requirements of 3.10, the casting shall be rejected.

4.4.3.2 Physical properties

4.4.3.2.1 High temperature. A random sample shall be selected from each inspection sublot or skid and tested for the maximum compression load and deformation as specified in 4.5.3. The sample for these tests shall be selected from the top of the casting. If the random sample fails to meet the requirements of 3.9.1, the inspection sublot or skid shall be rejected.

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4.4.3.2.2 Low temperature. A random sample shall be selected from each inspection sublot or skid and tested for the maximum compression load and deformation as specified in 4.5.3. The sample for these tests shall be selected from the top of the casting. If the random sample fails to meet the requirements of 3.9.2, the inspection sublot or skid shall be rejected.

4.4.4 Width of cut. The width of cut taken in splitting the grain shall not exceed the drawing requirement. For verification, the width of cut shall be inspected at the beginning of each 8 hours of production by making a partial cut of the grain.

4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with Paragraph 4.4.2.1. The Contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see Section 6 of MIL-A-48078 and 6.3 herein).

4.5 Test methods and procedures.

4.5.1 Composition.

4.5.1.1 Nitrocellulose. The nitrocellulose content shall be determined in accordance with MIL-STD-286, Method 209.2 or Method 209.6.

4.5.1.2 2-nitrodiphenylamine. The 2-nitrodiphenylamine content shall be determined in accordance with MIL-STD-286, Method 218.4.

4.5.1.3 Nitroglycerin. The nitroglycerin content shall be determined in accordance with MIL-STD-286, Method 208-1.

4.5.1.4 Triacetin. The triacetin content shall be determined in accordance with MIL-STD-286, Method 222.1.

4.5.1.5 Lead salicylate and lead 2-ethyl hexoate. Total lead content shall be determined in accordance with MIL-STD-286, Method 316.1; lead salicylate content shall be determined in accordance with MIL-STD-286, Method 225.1; lead 2-ethyl hexoate shall be determined by difference.

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4.5.2 Taliani test. This test shall be performed in accordance with MIL-STD-286, Method 406.1

4.5.3 Physical properties. Each specimen and specimen support shall conform to the requirements of Figure 1. Testing shall be conducted with approved test fixtures and temperature conditioning box. If the temperature conditioning box is not an integral part of the test fixture, transfer of the specimen from the temperature conditioning box to the test fixture shall be accomplished within a maximum of two minutes at high temperature and one minute at low temperature. The specimen shall be inserted between the fixture posts with steel cylinders the same diameter as the specimen and placed above and below the specimen forming one continuous concentric cylinder. The specimen shall be conditioned at the required temperature for three hours minimum and shall be tested at a crosshead speed of 2 inches per minute.

4.5.4 Strand burning rate. The strand burning rate shall be determined in accordance with Method 803.1 of MIL-STD-286 for test in nitrogen (-40°F) and a Government approved Contractor method for testing in oil (SAE #10, for 70°F and 140°F tests). The strands tested in oil shall be uninhibited.

4.5.5 Heat of explosion. This test shall be performed in accordance with MIL-STD-286, Method 802.1.

4.5.6 Radiographic inspection. The split finish machined propellant grain shall be radiographed with the beam parallel to the longitudinal axes of the grain. Observations shall be made for cracks, voids, fissures and foreign material in accordance with MIL-STD-453. A penetrameter or sample test block showing the types of defects for rejection shall be designed to assure a 2 percent minimum sensitivity.

4.5.7 Density. The density of the propellant grain shall be determined in accordance with OD19477, (Density Determination, Mercury Method, Casting Powder) by Department of the Navy, Bureau of Naval Weapons dated 19 May 1961 (see 6.7).

5. PACKAGING

5.1 Packing. Eight grains shall be packed in a fiber drum meeting the requirements of DOT specifications 21C of the Code of Federal Regulations Title 49 Parts 100 to 199.

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The drum wall shall be lined with a kraft backed commercial grade single faced corrugated fiber board using as many layers as required, for a tight pack. The drum size shall be the minimum required to package eight (8) propellant grains packed in two (2) layers of four (4) per layer. A barrier meeting the requirements of MIL-B-117, Type I, Class E, Style 1 shall be placed inside the lined drum. The barrier bag size shall be inserted into a spiral or convolute wound chipboard tube having the following dimensions: Inside diameter 6.865 + .015; outside diameter 7.190 - .015; length 6.050 - .059 inches. The tube winding shall be secured with dextrin or polyvinyl adhesive and the inside wall surface shall be smooth. Each grain shall be placed in the fiber drum with the inhibitor end up. A separator shall be placed between layers and on top of final layer. The separator shall be corrugated cardboard liners or separators, double-faced 'B' or 'C' Flute, 200-pound test board. The bag shall be folded and sealed with tape meeting the requirements of PPP-T-60. Filler constructed of the same material as the separators shall be placed on top of the contents as required to provide a tight pack. The fiber drum cover shall be sealed with 2-inch minimum wide tape meeting the requirements of PPP-T-76. All packing material inside the barrier bag shall not exceed 7 percent maximum moisture content.

5.2 Marking. The fiber drum shall be marked as follows:

DOT marking on top - "PROPELLANT EXPLOSIVES (SOLID) CLASS B"

On sidewall - "8 GRAIN, PROPELLANT, f/RAP 8-In. M753

DATE PACKED (MO/YR)
SUBLOT NO (INSERT NO)
SERIAL NUMBER (S)
WT (INSERT WT) CU (INSERT VOLUME)"

Marking shall be with black waterproof ink and lettering shall be 1/4 inch minimum height.

6. NOTES

6.1 Intended use. The item covered by this specification is intended for use in Rocket motor/Bulkhead Assemblies for M753 8 Inch Projectiles.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Surveillance samples required (see 3.13).
c. First article inspection, when required (see 3.14).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
(a) 3.11	Data feedback	DI-R-1724
(b) 3.12	Data cards	DI-L-1410
(c) 4.4.5	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Base grain powder test. In the event AHH base grain powder has been in storage for one year or longer, the powder shall be requalified in accordance with MIL-P-60432.

6.5 Submission of test data. In addition to the normal distribution of records, one copy of all test data will be forwarded to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-R and one copy to ATTN: DRDAR-LCN-DP, Dover, New Jersey 07801.

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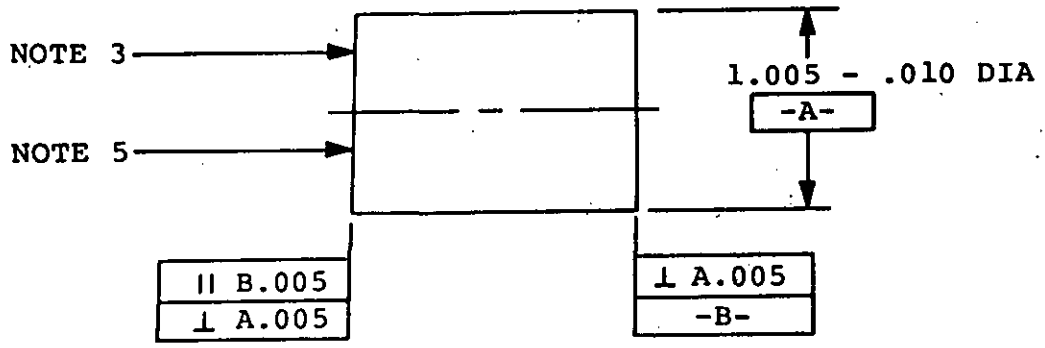
6.6 Surveillance sample. Samples shall be forwarded to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-LCE-M, Dover, New Jersey 07801, for the propellant surveillance test program in accordance with MIL-STD-652. This sample shall contain propellant pieces not exceeding one inch in diameter which may include machine turnings, broken tensile specimens, or pieces of grain segments, but no inhibitor materials are to be included.

6.7 Density measurements (OD19477). Copies of OD19477 (Density Determination, Mercury Method, Casting Powder) by the Department of the Navy, Bureau of Naval Weapons dated 19 May 1961 may be obtained from Commander U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801.

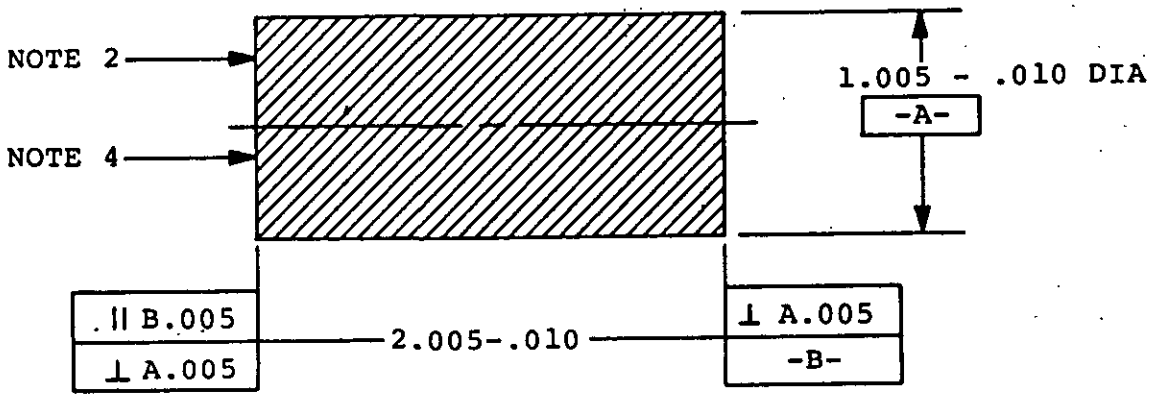
6.8 Supersession data. This specification includes the requirements of Purchase Description AR-PD-12, Revision A dated 11 April 1980.

6.9 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading US Army Armament Research and Development Command may also include dwgs. and specifications prepared by and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs and specifications. Technical data originally prepared by these activities are now under the cognizance of ARRADCOM.

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SPECIMEN SUPPORTS



GRAIN SPECIMENS

NOTES

1. MIL-A-2550 applies.
2. Specimen material is solid propellant.
3. Two steel specimen supports are required and shall be placed above and below the specimen prior to test.
4. 125 surface finish all over.
5. 125 surface finish on propellant surface.

FIGURE 1. Grain specimens and specimen supports for compression test.

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Preparing Activity

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(Project 1110-A158)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-63159 (AR) Propellant Grain M753 8 Inch Projectile

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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